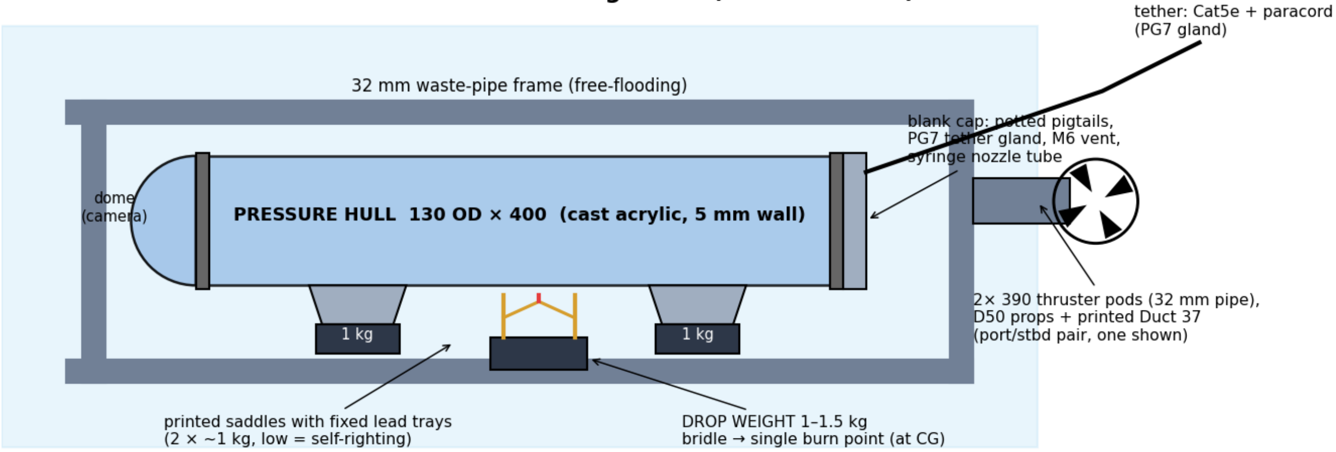
NEPTUNE — Mechanical & Electrical Integration

Master fit-and-connect reference: hull, sled, ballast, power, signals. Companion to the camera, navigation, blackbox and drop-weight release specs. Values marked ~ are settled at the bathtub ceremony (§8).

1 · General arrangement

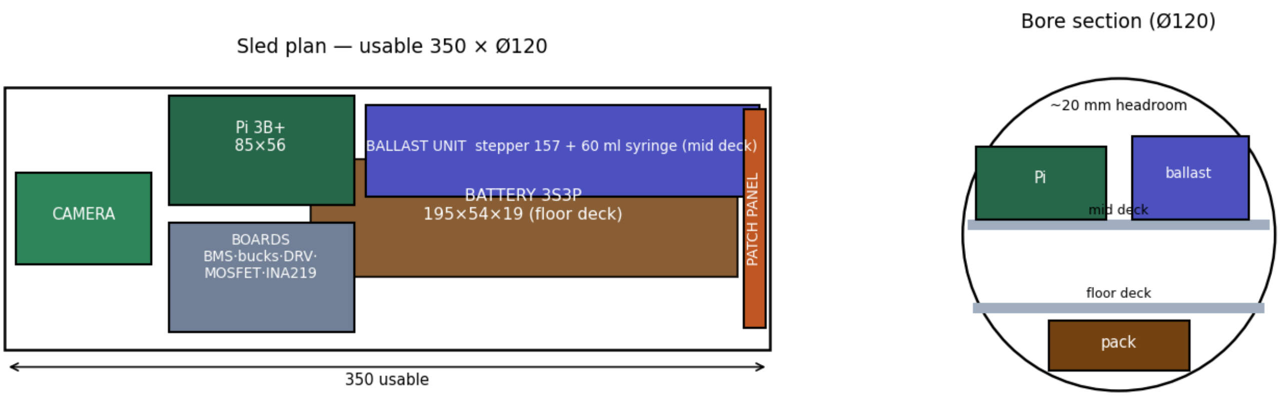
FIG 1 — General arrangement (side elevation)



Free-flooding 32 mm pipe frame carries: pressure hull in two printed saddles (fixed lead trays beneath), drop-weight seat between them at the CG, thruster pods aft (port/starboard pair), tether strain-relieved to the frame — **paracord takes tension, never the Cat5e or the gland**. Dome forward, all penetrations aft in the blank cap.

2 · Hull & sled

FIG 2 — Electronics sled (plan of both decks + bore section)

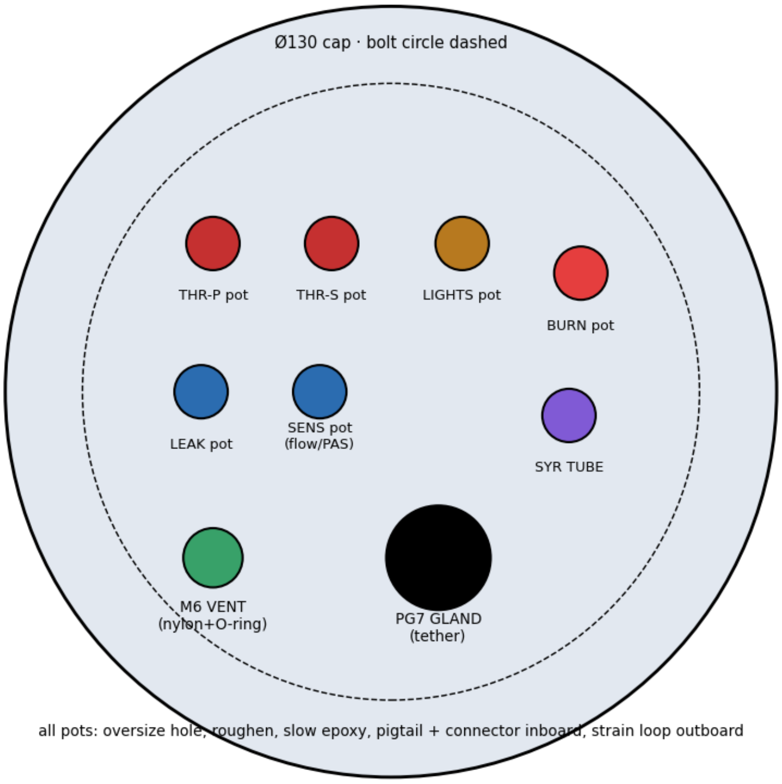


130 OD × 400 cast acrylic (ID ~120, usable length ~350 after flanges). Two-storey sled: pack on the floor deck (rear-biased, lowest mass = roll stability), camera at the nose near the dome's centre of curvature, Pi and ballast unit side-by-side on the mid deck, boards on the mid-deck forward section, patch panel at the rear face. Sled pulls out as one unit after unplugging the panel; leak sensor mounts to the HULL at the low point, not the sled.

Rules: power looms one rail, signal the other; service loops at every deck; one mid-sled support ring if sag appears; desiccant sachet + vent-plug crack before every cap pull.

3 · Rear cap penetrations

FIG 3 — Blank rear cap: penetration map (view from inside)

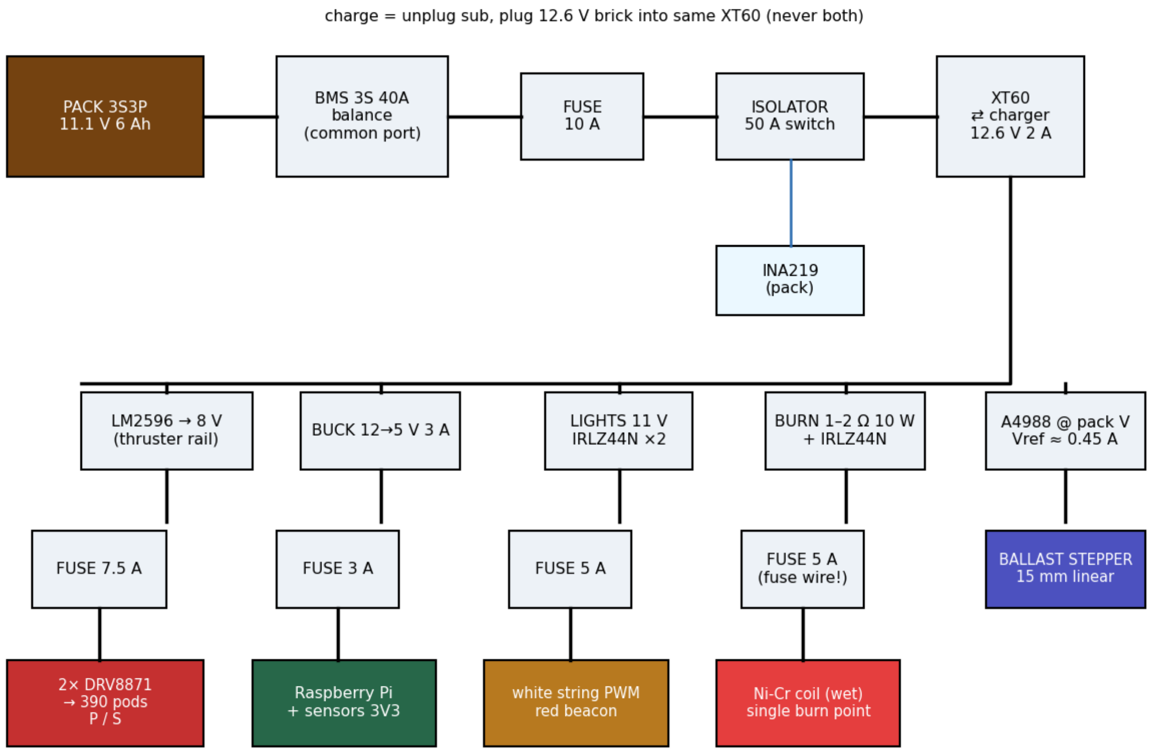


#	Penetration	Type	Wires/bore	Inboard end
1	Thruster P	potted epoxy	2× 18 AWG	XT30
2	Thruster S	potted epoxy	2× 18 AWG	XT30
3	Lights	potted epoxy	3× 20 AWG (W+, R+, GND)	XT30
4	Burn wire	potted epoxy	2× 18 AWG	JST-XH 2p
5	Leak sensor	potted epoxy	2× 28 AWG	JST-XH 2p
6	Flow / PAS	potted epoxy	4× 28 AWG	JST-XH 4p
7	Syringe nozzle	potted tube	4 mm OD silicone/PU tube	barb to syringe
8	Tether	PG7 gland	flat Cat5e (taped round)	RJ45 coupler
9	Vent / vac test	M6 nylon + O-ring	—	—

Every pot: pigtail + connector inboard (sled detaches without soldering), strain loop outboard, bench vacuum-test the finished cap over a jar before first assembly.

4 · Power distribution

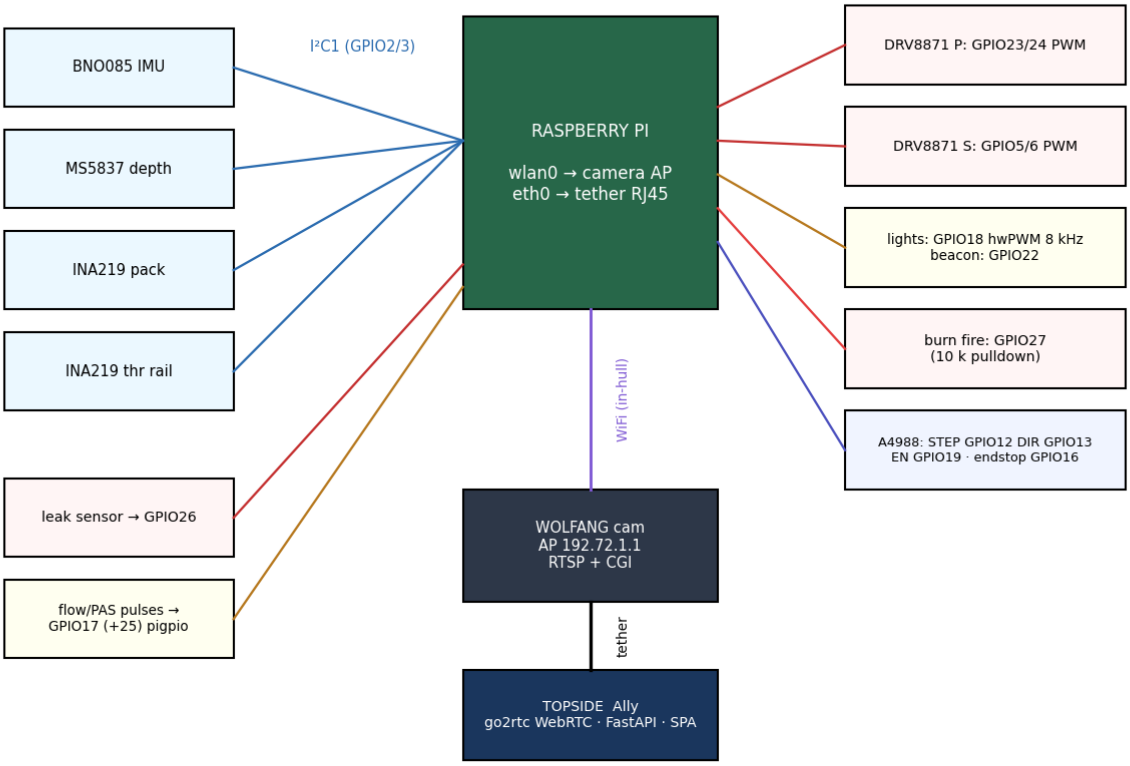
FIG 4 — Power distribution



Rail	Source	Fuse	Worst case	Notes
Thrusters	LM2596 @ 8 V	7.5 A	2× ~3 A run (390s)	DRV8871 limits stall @3.6 A; BTS7960 = upgrade path
Pi + sensors	5 V 3 A buck	3 A	~1.5 A	own regulator = immune to pack sag events
Lights	pack 11 V	5 A	~1 A	GPIO18 hwPWM 8 kHz; AWB toggle tied to state
Burn wire	pack + 1–2 Ω	5 A	2–3 A pulse	pulse 1.5 s, check, retry; log pack sag as confirmation
Ballast stepper	A4988 @ pack	(Pi rail logic)	~0.45 A/ph	Vref set LOW — heat in sealed hull
PACK TOTAL	3S3P 6 Ah	10 A main	~8 A abs.	INA219 on pack (+ optional 2nd on thruster rail)

5 · Signals & data

FIG 5 — Signal & data map

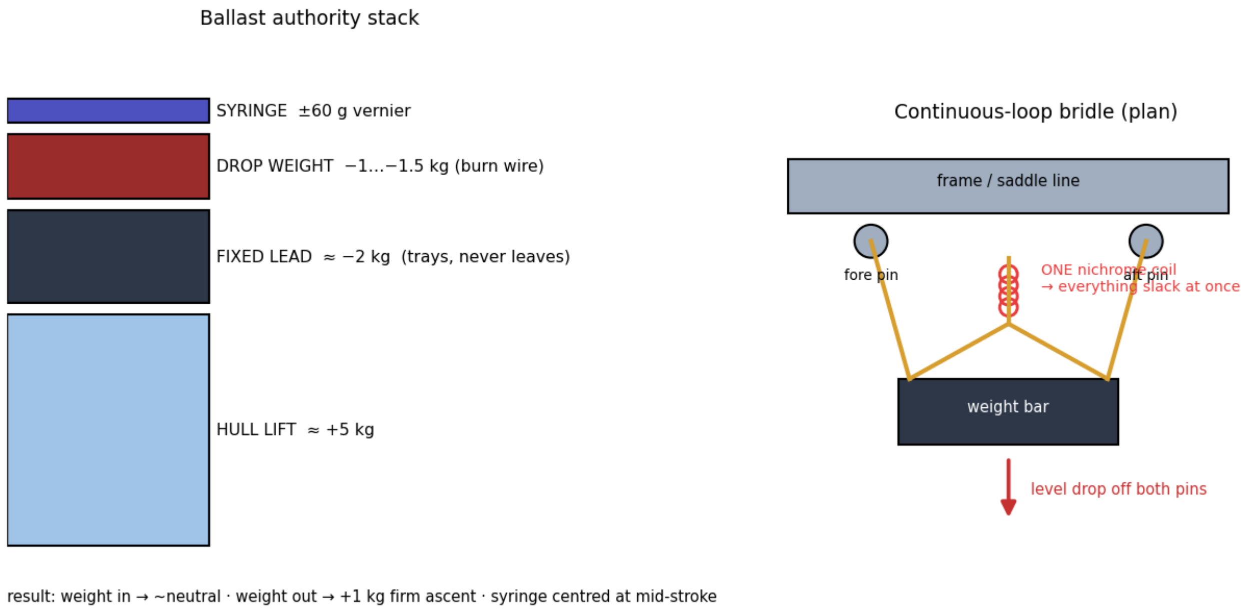


GPIO	Function	Dir	Notes
2 / 3	I²C1 SDA/SCL — BNO085, MS5837, INA219 ×2	bus	addresses must not clash; INA219 A0/A1 straps
17	flow sensor pulse	in	pigpio edge timestamps; period math
25	PAS second hall (direction)	in	quadrature with 17 if PAS fitted
26	leak sensor	in	interrupt → alarm + auto-surface policy
23 / 24	DRV8871 port IN1/IN2	out PWM	pigpio PWM
5 / 6	DRV8871 stbd IN1/IN2	out PWM	
18	white lights	out hwPWM	8 kHz — above camera banding
22	red beacon	out	surface-routine blink 0.2/1.8 s
27	burn-wire fire	out	10 k pulldown — no float-fire at boot
12 / 13 / 19	A4988 STEP / DIR / EN	out	EN idle-high = coils off = cool
16	syringe endstop (micro switch)	in	home position for ml-accurate trim

Data paths: camera → Pi over in-hull WiFi (AP 192.72.1.1, RTSP remux via go2rtc, CGI control serialised behind one lock); Pi → topside over tether Ethernet (WebRTC video, REST + WS telemetry, client-log upload). Default route stays on eth0; 192.72.1.1 pinned to wlan0.

6 · Ballast & release

FIG 6 — Ballast stack & single-point bridle release



Three independent layers: fixed trays cancel hull lift to slightly positive; drop weight biases diveable and is the emergency ascent (continuous-loop bridle, ONE burn point at CG, level release off both pins); syringe is the ±60 g vernier, centred at mid-stroke when the weight is seated. Escalation order on SURFACE: syringe full-positive → thrusters up → burn wire → (mk2) air blow. All logged with c_ids.

7 · Mass & buoyancy budget (pre-bathtub estimate)

Item	kg (– sinks / + floats)
Hull 130×400 + caps + dome, air-filled	+4.6 … +5.0
Frame, pods, fittings (flooded)	+0.2 … +0.4
Internals (pack 0.45, sled, Pi, camera, wiring)	–1.2 … –1.5
Motors, couplers, external sensors, hardware	–0.5 … –0.8
FIXED LEAD (trays, adjust at bathtub)	≈ –2.0 → boat barely positive
DROP WEIGHT	–1.0 … –1.5 → diveable
Syringe authority	±0.06

8 · Bathtub ceremony (sign-off procedure)

1. Full assembly, drop weight OUT, syringe mid-stroke, no fixed lead: record float attitude.

2. Add wheel-weight slices to trays until level and just-awash positive (~+150 g). Record per-tray mass and position.

3. Seat drop weight: boat should hold shallow neutral; trim residual with syringe. Record stroke position.

4. Fire syringe full range: confirm it moves the boat through neutral both ways.

5. Pull vent, vacuum test 10 min, reseal. Weighted overnight bucket soak with paper towel telltale.

6. Burn-wire bucket test with the flight bridle: level drop < 2 s. Re-arm, repeat ×3.

7. Record final numbers into the config + blackbox session_start.

Anything that later changes mass (new sensor, bigger prop, mud magnet) = re-run steps 2–4.